

Wi-Fi Security — Complete Preparation

Comprehensive Study Guide: Cryptographic Algorithms, Authentication Frameworks (802.1X / EAP / RADIUS), Enterprise vs. Personal Topologies, Handshake Mechanics & 25 Solved MCQs

Level 1 — Wi-Fi Security Protocols

1. WEP → RC4

WEP stands for **Wired Equivalent Privacy**. It was an early Wi-Fi security protocol designed to provide:

- Encryption
- Basic authentication
- Protection of wireless data over radio frequency transmission

WEP used the **RC4** stream cipher algorithm.

```
WEP
↓
RC4
↓
Encryption
```

Why is WEP insecure?

A major vulnerability in WEP was its use of a small **24-bit Initialization Vector (IV)** concatenated with the secret key. Because 24 bits allows only $2^{24} \approx 16.7 \text{ million}$ combinations, IVs repeat rapidly on high-traffic networks. Repeated IVs combined with cryptographic weaknesses in RC4 key scheduling allow attackers to mathematically recover the secret key from captured packets.

Conclusion: WEP is obsolete, deprecated by IEEE in 2004, and must never be used.

Accenture shortcut:

If you see: *Which encryption algorithm is associated with WEP?*

Answer → RC4

2. WPA → TKIP

WPA stands for **Wi-Fi Protected Access**. WPA was introduced as an interim patch to fix major security flaws in WEP while permitting legacy hardware to be upgraded via firmware without requiring hardware replacements.

WPA introduced **TKIP (Temporal Key Integrity Protocol)**.

Key Mechanisms of TKIP:

- **Dynamic key handling:** Generates unique keys per packet.
- **Per-packet key mixing:** Blends the root key with the transmitter MAC address and packet sequence number.

- **Replay protection:** Uses an extended 48-bit IV/sequence counter.
- **Message Integrity Checking:** Incorporates the *Michael* cryptographic check to prevent frame tampering.

WPA
↓
TKIP
↓
Improved wireless protection

Important: TKIP is not the same thing as AES.

For exam purposes:

- WPA → TKIP
- WPA2 → AES + CCMP

Accenture trap:

Question: Which protocol is primarily associated with WPA?

A. AES | B. TKIP | C. SAE | D. CCMP

Answer: B. TKIP

3. WPA2 → AES + CCMP ★★★★★ HIGH PRIORITY

WPA2 is the successor to WPA and provides substantially stronger, enterprise-grade security compliant with the full IEEE 802.11i standard. Its standard data-protection mechanism is **CCMP using AES**.

WPA2
↓
CCMP
↓
AES

What is AES?

AES (Advanced Encryption Standard) is the underlying symmetric block cipher that encrypts and decrypts wireless data payloads.

What is CCMP?

CCMP stands for **Counter Mode with Cipher Block Chaining Message Authentication Code Protocol**. CCMP is the security encapsulation protocol/mode designed around AES in WPA2. It provides both confidentiality (AES counter mode) and message integrity/authenticity (CBC-MAC).

Simple Analogy: Imagine sending a locked package.

- **AES** = The heavy-duty lock / encryption mechanism.
- **CCMP** = The complete armored courier protocol (locking, verification seal, anti-tamper checking).

Accenture shortcut:

- WPA2 uses which encryption technology? → **AES**
- WPA2 uses which protocol for data protection? → **CCMP**
- If an option says **AES + CCMP**, that represents the most complete and correct answer.

4. WPA3 → SAE ★★★★★ HIGH PRIORITY

WPA3 is the modern generation of Wi-Fi security (ratified 2018). For **WPA3-Personal**, the pivotal authentication improvement is the elimination of PSK in favor of **SAE**.

SAE (Simultaneous Authentication of Equals), based on the Dragonfly handshake, is a password-authenticated key exchange (PAKE) protocol that provides:

- Resistance against offline dictionary and brute-force password-guessing attacks.
- **Forward Secrecy:** Even if the password is breached later, past captured network sessions cannot be decrypted.
- Resilience against passive eavesdropping.

WPA2-Personal	→	PSK
WPA3-Personal	→	SAE

Accenture shortcut: WPA3-Personal uses what authentication protocol? → **SAE**

Level 1 — Complete Architectural Summary

Protocol	Cipher / Protocol Mode	Authentication Method	Status
WEP	RC4 (Stream Cipher)	Open / Shared Key (Static 24-bit IV)	BROKEN / OBSOLETE
WPA	TKIP (Temporal Key Integrity Protocol)	PSK / 802.1X	LEGACY / DEPRECATED
WPA2	AES + CCMP	PSK / 802.1X (EAP)	STRONG STANDARD
WPA3	AES-GCM (128/256-bit)	SAE (Simultaneous Authentication of Equals)	MODERN / HIGH SECURITY

One-Line Memory Trick:
WEP-RC4 | WPA-TKIP | WPA2-AES/CCMP | WPA3-SAE

Level 2 — Authentication Architecture

It is vital to distinguish between two foundational questions in wireless networking:

- **Level 1:** *How is wireless payload data encrypted and protected over the air?*
- **Level 2:** *How does the network verify identity and grant access permission?*

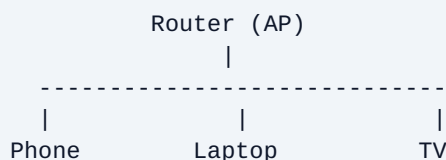
5. WPA2-Personal → PSK

WPA2-Personal is designed for home and small office (SOHO) networks. It uses a **Pre-Shared Key (PSK)**, commonly known as the Wi-Fi password.

User → Wi-Fi Password / Shared Secret → WPA2-Personal → Authentication → Network Access

Why "Pre-Shared"? Because the secret key is pre-configured and shared among all authorized devices and the Access Point without requiring an external server.

6. Why is WPA2-Personal suitable for homes?



In a home setup, configuring individual user accounts for every family member and smart device is unnecessary overhead. A single shared secret suffices.

7. WPA2-Enterprise → 802.1X

In an organization with hundreds or thousands of employees, a shared PSK fails because:

- When an employee departs, the entire network password must be changed across thousands of endpoints.
- Shared secrets prevent granular accountability and individual session auditing.

WPA2-Enterprise solves this by using **IEEE 802.1X** for port-based, individual authentication.

Employee Device → 802.1X → Individual Authentication → Dynamic Network Access

8. What is 802.1X?

802.1X = Port-Based Network Access Control. It keeps the network switch/AP port locked to unauthorized data traffic until identity validation succeeds. It defines three distinct architectural roles:

- **Supplicant:** The client software/device requesting network access (e.g., employee's laptop/phone).
- **Authenticator:** The network boundary device controlling physical/wireless port state (e.g., Access Point or Managed Switch).
- **Authentication Server:** The centralized identity server validating credentials (commonly a **RADIUS** server).

9. 802.1X → EAP

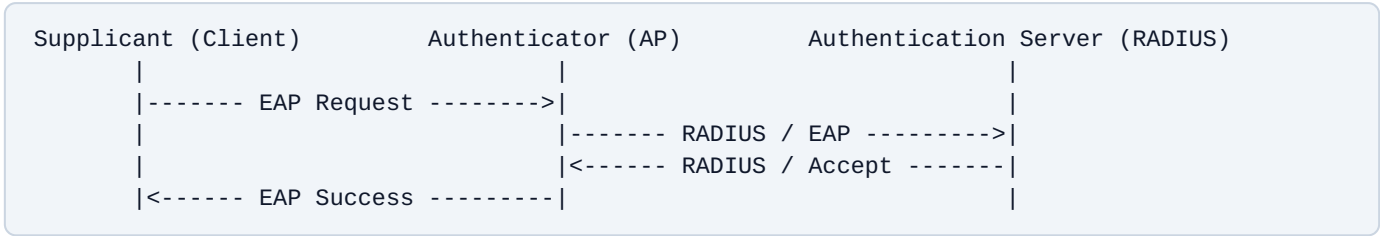
EAP (Extensible Authentication Protocol) is an authentication framework that carries diverse credential formats across 802.1X links. Common methods include **EAP-TLS** (certificates), **PEAP**, and **EAP-TTLS** (tunneled credentials).

Rule of thumb: 802.1X acts as the carrier framework, and EAP delivers the authentication payload.

10. EAP vs. RADIUS

EAP and RADIUS are fundamentally different entities in enterprise architecture:

- **EAP:** An extensible authentication protocol framework between the Supplicant and Server.
- **RADIUS:** The backend server and protocol executing user verification against enterprise directory databases.

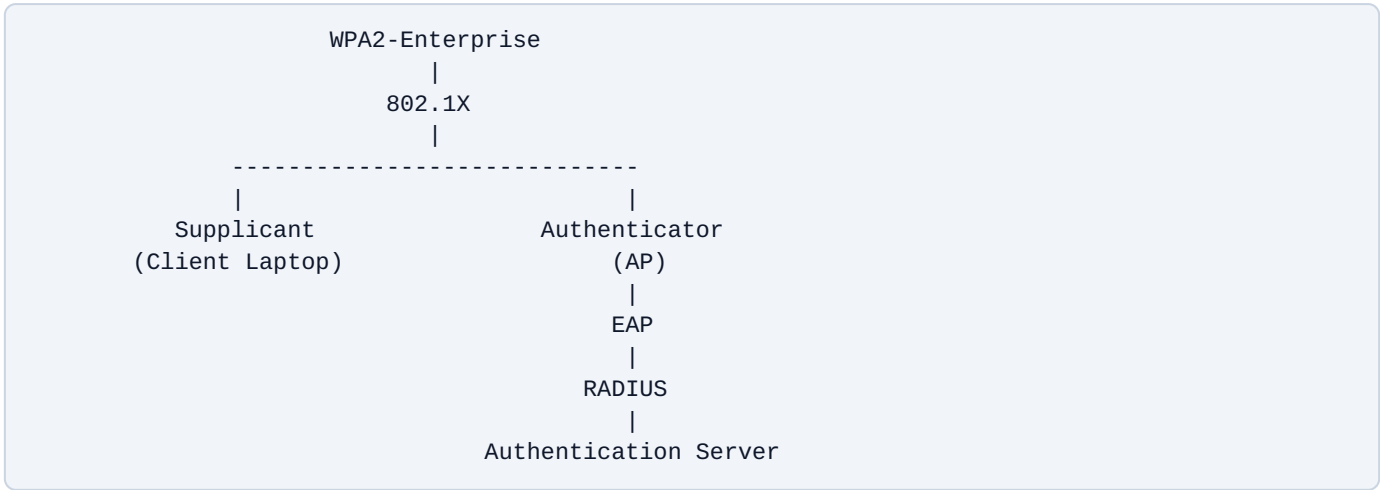


11. What does RADIUS do?

RADIUS (Remote Authentication Dial-In User Service) provides centralized AAA services:

- **Authentication:** *Who are you?* (Verifying credentials).
- **Authorization:** *What are you allowed to access?* (Assigning VLANs, ACLs, bandwidth profiles).
- **Accounting:** *What did you do?* (Logging session duration, login times, and bandwidth usage).

12. Complete Enterprise Authentication Flow ★★★★★ HIGH PRIORITY



Simplified Chain:
Client → 802.1X → EAP → RADIUS → Authentication Server Database

13. WPA2-Personal vs. WPA2-Enterprise Comparison

Feature	WPA2-Personal	WPA2-Enterprise
Authentication	PSK (Pre-Shared Key)	802.1X / EAP
Credentials	Single shared passphrase	Individual user credentials / certificates
RADIUS Server	Not required (Local AP validation)	Required (Centralized AAA server)
Deployment	Home / Small Office (SOHO)	Large Enterprise, University, Corporate
Scalability	Low (Key updates require mass change)	High (Centralized user directory)
User Revocation	Difficult (Must update all devices)	Instant (Disable individual user account)

Memory Trick:

- **PERSONAL** → Password / PSK
- **ENTERPRISE** → Individual Authentication / 802.1X / EAP / RADIUS

14. Key Distinction: Data Encryption vs. Authentication

Data Encryption & Integrity (Level 1)	Authentication & Access Control (Level 2)
• WEP → RC4 • WPA → TKIP • WPA2 → AES + CCMP • WPA3 → SAE / AES-GCM	• Personal: PSK (Pre-Shared Key) • Enterprise: 802.1X → EAP → RADIUS (AAA)

Wi-Fi Security — Practice MCQs & Explanations

Q1. Theory ★★★★★

Which Wi-Fi security standard uses AES with CCMP as its primary data-protection mechanism?

- A. WEP
- B. WPA
- C. WPA2
- D. WPA3

✓ **Answer: C. WPA2**

Detailed Explanation:

- **A. WEP** ✗: Uses RC4 stream cipher; suffers from serious architectural IV flaws.
- **B. WPA** ✗: Primarily associated with TKIP.
- **C. WPA2** ✓: Standardized AES-CCMP for encryption and integrity verification.
- **D. WPA3** ✗: Uses SAE for authentication and modern GCMP/AES cipher suites.

🧠 **Quick Memory:** WEP → RC4 | WPA → TKIP | WPA2 → AES+CCMP | WPA3 → SAE

Q2. Scenario ★★★★★

A company is currently using WEP for its wireless network. The security team wants to replace it with a modern protocol. Which is the best choice?

- A. Continue using WEP but increase the password length
- B. Replace WEP with WPA3
- C. Enable SSID hiding
- D. Enable MAC filtering

✓ **Answer: B. Replace WEP with WPA3**

Detailed Explanation:

- **A ✗**: Increasing password length does not eliminate WEP's 24-bit IV collision weakness.
- **B ✓**: WPA3 provides modern, robust encryption and protects against dictionary attacks.
- **C ✗**: SSID hiding is merely security-through-obscurity; SSIDs are still transmitted in cleartext probe frames.
- **D ✗**: MAC filtering is easily bypassed by sniffing wireless traffic and spoofing allowed MAC addresses.

🔗 **Exam Trick:** When replacing obsolete WEP with the strongest modern standard, choose WPA3.

Q3. Theory ★★★★★

Which encryption algorithm is associated with WEP?

- A. AES
- B. RC4
- C. RSA
- D. SHA-256

✓ **Answer: B. RC4**

Detailed Explanation:

- **A. AES ✗**: Symmetric block cipher used in WPA2/WPA3.
- **B. RC4 ✓**: Stream cipher utilized in the original WEP protocol.
- **C. RSA ✗**: Asymmetric public-key cipher used in TLS/certificates, not WEP bulk encryption.
- **D. SHA-256 ✗**: Cryptographic hash algorithm, not an encryption cipher.

Q4. Scenario ★★★★★

A student connects to a Wi-Fi network using WPA2-Personal. What authentication mechanism is most likely being used?

- A. SAE
- B. RADIUS only
- C. PSK
- D. Kerberos

✓ **Answer: C. PSK**

Detailed Explanation:

- **A. SAE** ✗: Used in WPA3-Personal.
- **B. RADIUS** ✗: Used in Enterprise deployments.
- **C. PSK** ✓: WPA2-Personal relies on Pre-Shared Key (password) authentication.
- **D. Kerberos** ✗: Network authentication protocol used in Windows Active Directory, not standard Wi-Fi personal access.

🧠 **Memory:** WPA2-Personal → PSK | WPA3-Personal → SAE

Q5. Scenario ★★★★★

An organization has 2,000 employees. Management does not want everyone to use the same Wi-Fi password. Which solution is most appropriate?

- A. WPA2-Personal
- B. WPA2-Enterprise
- C. WEP
- D. Hidden SSID

✓ **Answer: B. WPA2-Enterprise**

Detailed Explanation: Enterprise networks require individual credentialing. WPA2-Enterprise routes requests through 802.1X / EAP / RADIUS, allowing instant revocation when an employee leaves without disrupting other staff.

Q6. Theory ★★★★★

Which protocol is commonly used as the authentication server/backend in enterprise Wi-Fi?

- A. RADIUS
- B. ARP
- C. DNS
- D. DHCP

✓ **Answer: A. RADIUS**

Detailed Explanation: RADIUS provides centralized AAA (Authentication, Authorization, Accounting). ARP translates IP to MAC, DNS resolves hostnames, and DHCP assigns IP configuration parameters.

Q7. Scenario ★★★★★

An employee connects a laptop to an enterprise Wi-Fi network. The network uses 802.1X. Which component acts as the supplicant?

- A. RADIUS server
- B. Access Point
- C. Employee's laptop
- D. DNS server

✓ **Answer: C. Employee's laptop**

Detailed Explanation: In IEEE 802.1X architecture:

- **Supplicant:** The client endpoint device (Employee laptop/phone).
- **Authenticator:** The Access Point or Switch controlling network entry.
- **Authentication Server:** The RADIUS backend server.

Q8. Theory ★★★★★

What is the primary purpose of 802.1X?

- A. Encrypting files using AES
- B. Providing port-based network access control
- C. Translating domain names
- D. Assigning IP addresses

✓ **Answer: B. Providing port-based network access control**

Detailed Explanation: IEEE 802.1X standardizes Port-Based Network Access Control (PNAC), establishing a gatekeeping mechanism between the client and the underlying network.

Q9. Scenario ★★★★★

You are configuring a new WPA3-Personal home router. Which authentication mechanism should you expect?

- A. WEP
- B. TKIP
- C. SAE
- D. RADIUS

✓ **Answer: C. SAE**

Detailed Explanation: WPA3-Personal replaces legacy PSK authentication with SAE (Simultaneous Authentication of Equals) to protect against dictionary attacks and offer forward secrecy.

Q10. Scenario ★★★★★

A security analyst captures a WPA2 4-way handshake from a wireless network. What does this not automatically provide to the attacker?

- A. Authentication exchange information
- B. Information useful for offline password guessing
- C. The plaintext Wi-Fi password directly
- D. Wireless handshake data

✓ **Answer: C. The plaintext Wi-Fi password directly**

Detailed Explanation: Capturing the 4-way handshake captures nonces and cryptographic message authentication codes (MICs), not the plaintext password. An attacker must perform offline dictionary or brute-force computation against the captured hash to deduce weak passwords.

🚫 **Exam Trap:** Handshake captured ≠ password immediately revealed in plaintext!

Q11. Theory ★★★★★

Which key is derived during the WPA2 4-way handshake for a particular client/AP session?

- A. SSID
- B. PTK
- C. MAC address
- D. PSK only

✓ **Answer: B. PTK**

Detailed Explanation: The Pairwise Transient Key (PTK) is dynamically computed during the 4-way handshake using:

PMK + ANonce (from AP) + SNonce (from Client) + AP MAC + Client MAC → PTK

PTK is used to encrypt unicast traffic and authenticate messages for that specific session.

Q12. Scenario ★★★★★

A coffee shop has a Wi-Fi network named CoffeeShop_Free. An attacker creates another Wi-Fi network with the same name and tries to convince customers to connect to it. What type of attack is this?

- A. Deauthentication attack
- B. Evil Twin attack
- C. MAC filtering
- D. ARP poisoning

✓ **Answer: B. Evil Twin attack**

Detailed Explanation: An Evil Twin is a rogue AP configured with identical SSID and security parameters to trick unsuspecting users into connecting, allowing the attacker to snoop traffic or deploy captive portal credential harvesters.

Q13. Scenario ★★★★★

An unauthorized wireless access point is discovered inside a company's network. What is this generally called?

- A. Rogue Access Point
- B. Evil Twin only
- C. WPA3
- D. SSID

✓ **Answer: A. Rogue Access Point**

Detailed Explanation: A Rogue AP is any unauthorized AP connected directly to an internal corporate network infrastructure without administrative knowledge or approval.

Q14. Scenario ★★★★★

A wireless attacker sends forged management frames that cause connected clients to disconnect from an AP. Which attack is most likely occurring?

- A. Evil Twin
- B. Deauthentication attack
- C. DNS poisoning
- D. Password spraying

✓ **Answer: B. Deauthentication attack**

Detailed Explanation: In standard 802.11 Wi-Fi, management frames (including deauthentication and disassociation frames) are unencrypted. Attackers spoof the AP's MAC address and send deauth frames to kick clients off the network (often to capture the 4-way handshake upon reconnection).

Q15. Theory ★★★★★

What does SSID identify?

- A. A Wi-Fi network's name/identifier
- B. The encryption algorithm
- C. The user's password
- D. The RADIUS server

✓ **Answer: A. A Wi-Fi network's name/identifier**

Detailed Explanation: SSID stands for **Service Set Identifier**. It is the human-readable string (up to 32 bytes) that identifies a specific 802.11 wireless local area network.

Q16. Scenario ★★★★★

A network administrator hides the SSID and claims: "Now attackers cannot discover or attack our Wi-Fi." Which statement is correct?

- A. Correct; hidden SSID provides complete security
- B. Correct; hidden SSID encrypts all traffic
- C. Incorrect; hiding SSID is not a substitute for strong authentication and encryption
- D. Incorrect; hiding SSID disables WPA2

✓ **Answer: C. Incorrect; hiding SSID is not a substitute for strong authentication and encryption**

Detailed Explanation: Disabling SSID beaconing is purely "security through obscurity". The SSID is still transmitted in plaintext whenever a valid client connects (in Probe Request/Response and Association packets). Packet sniffers can uncover hidden SSIDs within seconds.

Q17. Scenario ★★★

A router allows only certain MAC addresses to connect. The administrator believes this makes the network completely secure. What is the biggest problem with this assumption?

- A. MAC addresses cannot be changed
- B. MAC addresses can be spoofed
- C. MAC filtering automatically disables encryption
- D. MAC filtering requires RADIUS

✓ **Answer: B. MAC addresses can be spoofed**

Detailed Explanation: MAC addresses are transmitted unencrypted in 802.11 frame headers. Attackers can sniff legitimate MAC addresses over the air and clone/spoof them on their own network cards.

Q18. Scenario ★★★

A user wants the easiest possible method to connect a new device to a home router by pressing a physical button on the router. Which feature is likely being used?

- A. WPS
- B. RADIUS
- C. SAE
- D. CCMP

✓ **Answer: A. WPS**

Detailed Explanation: **WPS (Wi-Fi Protected Setup)** allows simplified device association via Push-Button Configuration (PBC) or an 8-digit PIN. (Note: WPS PIN mode is vulnerable to brute-force attacks via Reaver/Bully and is recommended to be disabled).

Q19. Theory ★★★★★

Which pairing is INCORRECT?

- A. WEP — RC4
- B. WPA — TKIP
- C. WPA2 — AES/CCMP
- D. WPA3-Personal — TKIP

✓ **Answer: D. WPA3-Personal — TKIP**

Detailed Explanation: Correct mappings are:

- WEP → RC4
- WPA → TKIP
- WPA2 → AES / CCMP
- WPA3-Personal → SAE (Simultaneous Authentication of Equals)

TKIP is an obsolete protocol tied to legacy WPA and has zero association with WPA3.

Q20. Scenario ★★★★★

A company wants an architecture where employee laptops authenticate individually via an Enterprise AP to a Central Authentication Server. Which combination is most appropriate?

- A. WPA2-Personal + PSK
- B. WPA2-Enterprise + 802.1X + EAP + RADIUS
- C. WEP + RC4
- D. WPA + TKIP

✓ **Answer: B. WPA2-Enterprise + 802.1X + EAP + RADIUS**

Detailed Explanation: Enterprise Wi-Fi connects Supplicants (Client Laptops) through Authenticator APs to backend AAA servers via IEEE 802.1X carrying EAP payloads over RADIUS.

Q21. Scenario (Tricky) ★★★★★

A company says: "We use WPA2-Enterprise, so RADIUS itself encrypts all Wi-Fi traffic." What is wrong with this statement?

- A. WPA2-Enterprise doesn't use authentication
- B. RADIUS is primarily part of the authentication/AAA infrastructure, while Wi-Fi data protection is handled by the wireless security mechanisms
- C. RADIUS is an encryption algorithm
- D. WPA2-Enterprise uses WEP

✓ **Answer: B. RADIUS is primarily part of the authentication/AAA infrastructure, while Wi-Fi data protection is handled by the wireless security mechanisms**

Detailed Explanation: Common conceptual trap! RADIUS authenticates the user and helps derive the Master Session Key (MSK). The actual over-the-air data encryption and integrity are executed locally between the client and AP using AES-CCMP.

Q22. Theory (Very Important) ★★★★★

Which of the following correctly represents the relationship?

- A. WPA2 → TKIP → RC4
- B. WPA → AES → CCMP
- C. WPA2 → CCMP → AES
- D. WPA3 → WEP → RC4

✓ **Answer: C. WPA2 → CCMP → AES**

Detailed Explanation: WPA2 utilizes CCMP as its encapsulation and integrity protocol mode, which in turn implements the AES block cipher.

Q23. Scenario (Enterprise) ★★★★★

A company wants to immediately revoke an employee's network access when the employee leaves the organization without changing credentials for every other employee. Which approach is better?

- A. Shared WPA2-PSK
- B. WPA2-Enterprise with individual authentication
- C. WEP
- D. Hidden SSID

✓ **Answer: B. WPA2-Enterprise with individual authentication**

Detailed Explanation: In enterprise mode, access permissions are bound to individual user accounts (or PKI certificates) in Active Directory/RADIUS. Disabling that single account instantly revokes access without affecting any other user.

Q24. Theory ★★★★★

Which of the following provides authentication, rather than simply identifying the Wi-Fi network?

- A. SSID
- B. PSK
- C. MAC address only
- D. Channel number

✓ **Answer: B. PSK**

Detailed Explanation:

- **SSID** ✗: Identifies network name only.
- **PSK** ✓: Pre-Shared Key performs cryptographic proof-of-identity authentication.
- **MAC address** ✗: Layer-2 hardware address (can be trivially spoofed).
- **Channel number** ✗: Physical radio frequency band.

Q25. Mixed Scenario (Accenture Level) ★★★★★

A company currently has: WEP, Shared password, Hidden SSID, and MAC filtering. The security team can make only one immediate change. Which change gives the biggest security improvement?

- A. Hide the SSID more effectively
- B. Add more MAC addresses to the filtering list
- C. Replace WEP with WPA3
- D. Change the Wi-Fi channel

✓ **Answer: C. Replace WEP with WPA3**

Detailed Explanation: The root vulnerability is WEP's fatally flawed RC4 encryption. Hiding SSIDs, adding MAC filters, or changing channels provides zero defense against passive cryptographic key extraction. Migrating directly from WEP to modern WPA3 replaces the broken cipher with robust AES-GCM and SAE authentication.

WEP (Broken RC4) → WPA3 (SAE / AES-GCM) = Maximum Security Jump